

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used for data collection.
Data analysis	All code used to generate results for this study, including data preprocessing, missing data handling, data analysis, visualization, and data modeling, is available on the author's GitHub page: https://github.com/kannyjyk/UKB-MDRMF. Software and Libraries Used Data analysis was performed using Python and R with the following versions: • Python: 3.7 or higher • R: 4.3 or higher Required Python Libraries: • lifelines==0.27.8 • torch==2.1.0 • numpy==1.26.2 • ipython==8.15.0 • matplotlib==3.7.2 • scikit-learn==1.2.2 • lightgbm==4.1.0 • xgboost==1.7.6 • pandas==2.1.1

- seaborn==0.13.2
- tqdm==4.66.1
- scipy==1.11.2
- fitz==1.23.20

Required R Packages:

- tidyverse (2.0.0)
- data.table (1.14.8)
- missRanger (2.4.0)
- scales (1.2.1)
- forcats (1.0.0)
- ggsci (3.0.0)
- ggpubr (0.6.0)
- optparse (1.7.3)
- plyr (1.8.9)
- zip (2.3.0)

All required dependencies can be installed using standard package management tools (pip for Python and install.packages() for R).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

This study was conducted using the UK Biobank Resource under Application Number 98327, with ethical approval from the North West Multi-Centre Research Ethics Committee (MREC) as a Research Tissue Bank (RTB). The UK Biobank study protocol is available online ([\url{https://www.ukbiobank.ac.uk/}](https://www.ukbiobank.ac.uk/)), and all participants provided written informed consent for data provision and linkage. UK Biobank is a large-scale biomedical resource comprising 502,467 individuals aged 37 to 72, recruited between 2006 and 2010 in the United Kingdom. The dataset includes demographic, biological, lifestyle, environmental, measurement, mental health, and imaging data.

To assess the generalizability of our model beyond UK Biobank, we performed external validation using the All of Us dataset, a diverse U.S. cohort designed to include underrepresented populations in biomedical research. The dataset integrates electronic medical records and participant surveys, with all participants providing informed consent. Ethical oversight is provided by the All of Us Institutional Review Board. This dataset was used for model validation, leveraging its broad demographic and clinical diversity to evaluate performance across populations.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	The study utilized data from the UK Biobank, which includes sex and gender information for all participants.
Reporting on race, ethnicity, or other socially relevant groupings	The UK Biobank dataset includes race and ethnicity information for almost all participants. This data was used to analyze the potential differences in disease prediction and risk assessments across various racial and ethnic groups, ensuring the model's applicability and fairness across diverse populations.
Population characteristics	The study population comprised participants from the UK Biobank, which includes over 502,467 individuals aged 37-72 at the time of recruitment. The dataset encompasses a wide range of demographic, lifestyle, and health-related information, making it representative of the general UK population within the specified age range.
Recruitment	Participants were recruited by the UK Biobank between 2006 and 2010 from across the United Kingdom. The recruitment process was designed to be inclusive and representative of the UK population.
Ethics oversight	The UK Biobank has obtained ethical approval from the North West Multi-centre Research Ethics Committee (MREC), which covers the UK. All participants provided informed consent, and the study was conducted in accordance with the Declaration of Helsinki. The ethical approval ensures that the participants' rights and confidentiality are protected.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	This study utilized the UK Biobank dataset, which includes data from over 500,000 participants. The large sample size provides strong statistical power and enhances the generalizability of the findings across diverse populations.
Data exclusions	Data exclusions were applied based on UK Biobank's quality control measures. Participants with missing or incomplete data for key variables were excluded from specific analyses to maintain the accuracy, reliability, and integrity of the results. Details of exclusion criteria are provided in the Methods section.
Replication	The analyses in this study are based on the UK Biobank dataset, a well-established and widely used resource in biomedical research. The methods and procedures are described in detail to ensure reproducibility, and the study can be replicated using the same dataset and analytical framework.
Randomization	As this study is observational and utilizes pre-existing UK Biobank data, randomization was not applicable. Participants were not assigned to specific groups, as the dataset was collected independently of this study.
Blinding	Blinding was not applicable, as this study involved a secondary analysis of pre-collected UK Biobank data. Investigators did not participate in data collection or group allocation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable to this study.
Novel plant genotypes	Not applicable to this study.
Authentication	Not applicable to this study.